

## GIS Developer & Analyst

I am a sociable, determined, and ambitious individual. I naturally navigate both formal and informal situations in any environment. I am adept at explaining complex concepts with extreme simplicity, tailoring my language to the audience—whether students or professionals at international scientific congresses.

Leveraging my expertise, I offer a variety of services, including professional cartography, characterization, quantification, and modeling of natural processes, ore deposit exploration, GIS software development, geospatial analysis, and remote sensing. Additionally, I provide scientific writing, management of experimental and laboratory activities, and professional sampling of environmental matrices. I specialize in hydrogeological instability and erosion phenomena in wildfire-affected areas.

## PROFESSIONAL EXPERIENCE

### Technical Internship

October 2023 – January 2024

CSIC-IGME - Madrid, Spain

- Remote Sensing
- Rockfall modeling with STONE in wildfire-affected areas

### Predoctoral Fellow

April 2022 - October 2022

University of Cagliari, Italy

- Characterization of geomorphological processes in wildfire-affected areas
- Soil sampling
- Final report

### Technical Internship

June 2019 - December 2019

ARPAS- Cagliari

- Determination of background values in abandoned mining areas (proUCL)
- Production of geochemical mapping and sampling preparation
- Final Report

### Technical Internship

May 2017 - November 2017

ARPAS - Cagliari

- Characterization of Radon risk exposure
- Sample preparation for XRD, XRF, and Gamma Spectrometry analysis
- Final Report

## EDUCATION

### PhD in Earth and Environment Sciences

October 2022 - February 2026

University of Cagliari

### Licensed Geologist

June 2021

University of Cagliari

### Degree in Geological Sciences and Technologies

September 2018 - July 2020

University of Cagliari

## LANGUAGES

|           |                        |
|-----------|------------------------|
| English   | Proficient, C1         |
| French    | Upper-Intermediate, B2 |
| Spanish   | Upper-Intermediate, B2 |
| Italian   | Native                 |
| Sardinian | Native                 |

## TECHNICAL SKILLS

**OS:** Windows, Linux e Android. **Programming:** Python & software development

**Productivity & Design:** Microsoft Office Suite, Open Source Office Suite, Inkscape, MS Teams

**GIS & Geospatial:** QGIS, ArcGIS, ArcPy, SUBSTR8. **Command Line:** shell & bash

## PUBLICATIONS

- Pala, C., Naitza, S., & Pioli, L. (2026). Stratigraphy and volcano-structural evolution of the Montiferru Volcanic Complex, Sardinia. Journal of Maps, 22(1). <https://doi.org/10.1080/17445647.2026.2632982>
- Pala, C., Melis, M.T., Pioli, L. et al. Sediment generation through thermal spalling during the 2021 montiferru planargia wildfire and its contribution to postfire debris flows. Sci Rep 15, 30918 (2025). <https://doi.org/10.1038/s41598-025-15527-2>
- Miranda, P., Pala, C., Mateos Ruiz, R. M., Galve, J. P., Millares, A., Martínez Corbella, M., ... & Sarro Trigueros, R. (2025). Evaluación de inestabilidades del terreno para la recuperación de áreas quemadas: el incendio de los Guájares (Granada).

## SOFTWARE DEVELOPMENT

- Pala, C. (2026). INUE Interactive and Userfriendly Emergency tool for postfire erosion susceptibility mapping (1.1 'άλφα'). Zenodo
- Pala, C. (2026). SUBSTR8 - An open platform for building GIS applications (1.0 Assíle). Zenodo. <https://doi.org/10.5281/zenodo.19462380>

## ADDITIONAL SKILLS

- Driving License (Category B) and own vehicle
- Assistance in drafting proposals for European Union project calls

I hereby authorize the processing of the personal data contained in this CV pursuant to Italian Legislative Decree 101/2018 and the GDPR (EU Regulation 2016/679).

Cuglieri, 15 April 2026

